

Exercise Sheet 9

Diffusion and Epidemics

5942UE Network Science

26 March 2026

Prof. Dr. Michael Granitzer

9.A Diffusion as a Network Process

Exercise 9.A.1: Simple Diffusion Trace

Directed graph: , , , .

Node is active at . At each step, an active node activates all out-neighbours.

1. Trace the active set at .
2. Which node is never reached? Why?
3. If you add the edge , does anything change for this seed?

Exercise 9.A.2: Structural Bottlenecks

Two 8-node graphs with seed :

- **Star:** connected to every other node.
 - **Line:**
1. How many steps until all nodes are active in each?
 2. Where is the structural bottleneck?
 3. A single edge fails. Which topology is more affected?

Exercise 9.A.3: Centrality and Diffusion Speed

Load the karate club graph and BFS-diffuse from the highest-degree node.

1. How many steps until all 34 nodes are active?
2. Repeat from the lowest-degree node. How does the curve change?
3. What does this say about hubs?

Exercise 9.A.4: Paths and Reachability

Company-wide email cascade from CEO () must reach 500 employees. The network has a core-periphery structure: 50 managers in a dense core, each linked to peripheral employees.

1. Does a message from to a peripheral employee require passing through the core?
2. If 5 of 50 managers are absent, what fraction of periphery is potentially unreachable?
3. How could the company become more resilient?

9.B Epidemic Models and Simple Contagion

Exercise 9.B.1: SIR Trace

SIR rules (synchronous): if any infected neighbour, with probability . at the next step (single-step infectious period).

Path graph with infected at , .

1. Trace each node's state at .
2. When does become infected?
3. With , what is ?

Exercise 9.B.2: Structural Effect of Topology

SIR with , one-step recovery, seed = node 1.

- **Line L:**
 - **Star S:** node 1 is the hub of
1. Maximum simultaneously infected in L?
 2. Maximum simultaneously infected in S?
 3. Which produces a sharper epidemic peak? Implications?

Exercise 9.B.3: SIR Simulation in NetworkX

Implement synchronous SIR on the karate club graph and analyse it.

1. At what step does the epidemic peak?
2. What fraction ends in (ever infected)?
3. Vary from 0.1 to 0.8. Where does the epidemic reliably infect ?

Exercise 9.B.4: Simple vs. Complex Contagion

For each scenario, decide whether it is **simple** (one contact sufficient) or **complex** (multiple confirmations needed):

(a) A cold virus. (b) Adopting a new software tool only after three colleagues recommend it. (c) Sharing a news article posted by one friend. (d) Joining a protest after pressure from multiple community members. (e) Updating a belief after seeing a claim from several independent sources.

For (b) and (d), how must network structure differ from simple contagion?

Wrap-Up

- diffusion is bounded by reachable paths; redundant paths don't help unless primary paths fail
- bottlenecks and hubs control epidemic dynamics
- the same produces dramatically different curves on different topologies
- complex contagion reverses the role of bridges and clusters compared to simple contagion